

ECA5612 –Advanced Wireless Communication Systems 5G / 6G Communication Systems.

Ex.No. 2: Implementing TDMA Slot Allocation in GSM. Date

Objective 1: Matlab Code and Output:

Program:

```
clc; clear; close all;
```

```
FrameTime = 4.615; Slots = 8;
```

```
SlotDuration = FrameTime/Slots;
```

```
t = 0:SlotDuration:FrameTime;
```

```
figure
```

```
subplot(2,1,1)
```

```
stem(t,ones(size(t)),'filled')
```

```
title('TDMA Time Slot Structure')
```

```
xlabel('Time (ms)')
```

```
ylabel('Slot Level')
```

```
grid on
```

```
subplot(2,1,2)
```

```
plot(t,ones(size(t)),'-o','LineWidth',1.5)
```

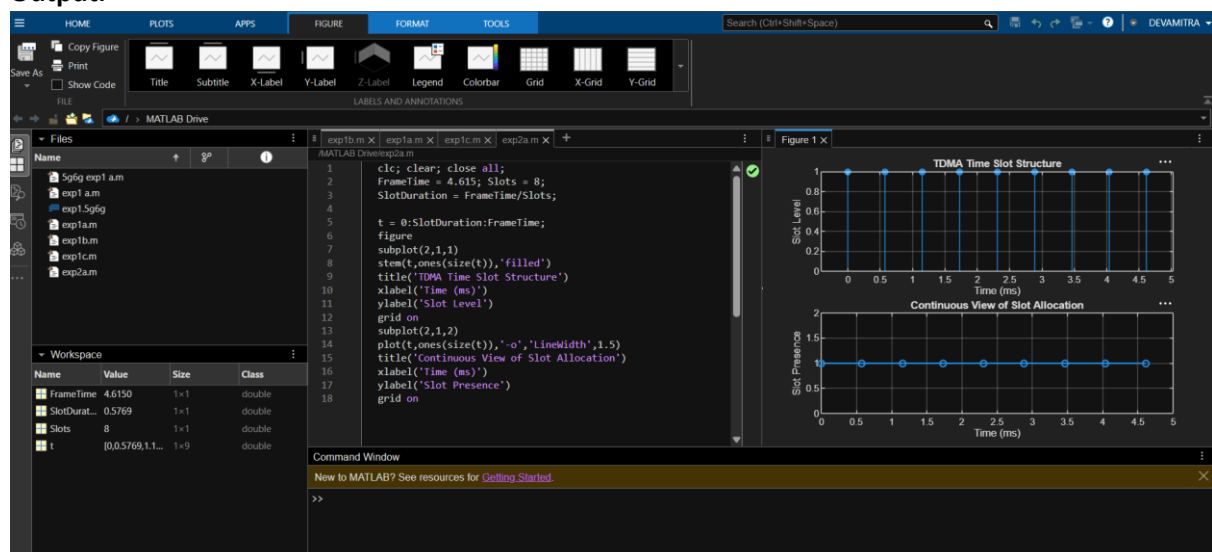
```
title('Continuous View of Slot Allocation')
```

```
xlabel('Time (ms)')
```

```
ylabel('Slot Presence')
```

```
grid on
```

Output:



Objective 2: Matlab Code and Output

Program:

```
clc; clear; close all;

TotalSlots = 8;

AllocatedSlots = 6;

FreeSlots = TotalSlots - AllocatedSlots;

Utilization = (AllocatedSlots/TotalSlots)*100;

disp('Frame Utilization (%)')

disp(Utilization)

figure

% Graph 1: Simple Line Plot (SAFE - no bar function)

subplot(2,1,1)

plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)

title('GSM Frame Utilization (Allocated vs Free)')

xlabel('Slot Type Index (1=Allocated, 2=Free)')

ylabel('Number of Slots')

grid on

% Graph 2: Pie Chart (SAFE)

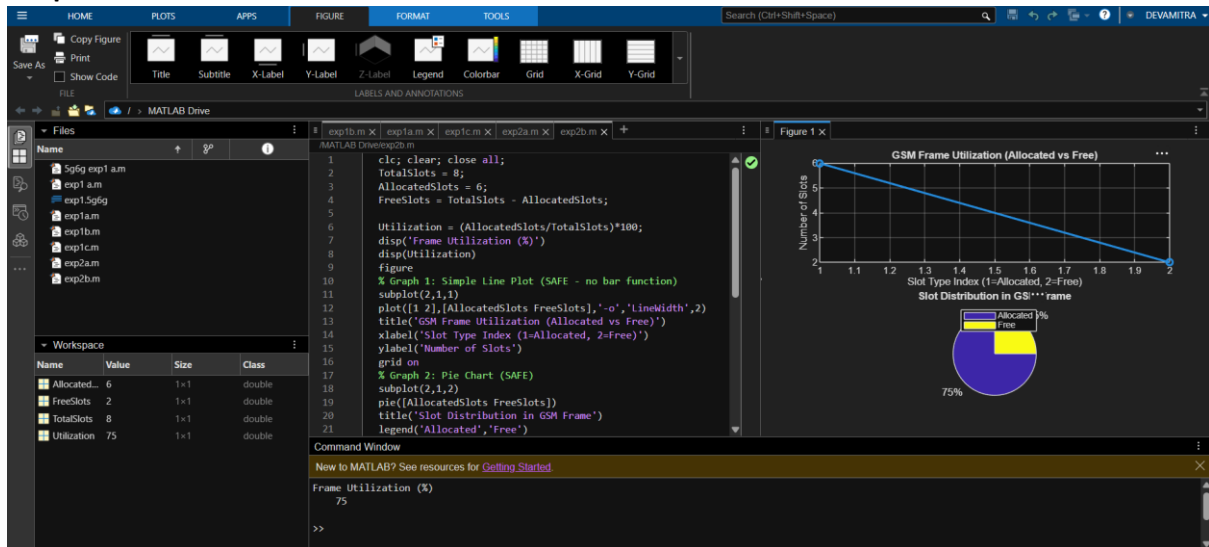
subplot(2,1,2)

pie([AllocatedSlots FreeSlots])

title('Slot Distribution in GSM Frame')

legend('Allocated','Free')
```

Output:



Objective 3: Matlab Code and Output

Program:

```
clc; clear; close all;

u = 1:8;

a = [ones(1,5) zeros(1,3)];

b = 1 - a;

disp(table(u,a,'VariableNames',{'User','Allocated'}))

subplot(2,1,1)

stem(u,a,'filled')

title('TDMA Slot Allocation')

xlabel('Users')

ylabel('Status')

grid on

subplot(2,1,2)

plot(u,a,'-o',u,b,'-s')

legend('Allocated','Blocked')

title('TDMA Slot Utilization')

xlabel('Users')
```

ylabel('Status')

grid on

Output:

